ECE 685 Final Projects

Project 1: Hypernetworks for Implicit Neural Representations

Required: 3 people

In this project, we study *implicit neural representations* (INRs) and explore how they can be generated using *hypernetworks*. Unlike conventional neural networks that operate on fixed discrete grids (such as pixels in an image), implicit neural representations model signals as *continuous functions*. This allows signals such as images, audio, or physical fields to be represented as coordinate-based neural networks.

The goal of this project is to investigate how neural networks can learn to generate such representations automatically. In particular, students will implement a *hypernetwork* that takes an image as input and produces the weights of a coordinate network that implicitly represents the image. Students will train and evaluate such models on CIFAR-10 and Celeb-A, and explore whether the learned representations can be useful for downstream tasks.

Implicit Neural Representations

An implicit neural representation models a signal as a function

$$f_\theta : \mathbb{R}^d \to \mathbb{R}^k,$$

where the input corresponds to a spatial coordinate and the output corresponds to the signal value at that coordinate. For example, an image can be represented by a neural network

$$f_\theta(x, y) \to (R, G, B),$$

which maps pixel coordinates (x, y) to color values. Instead of storing image pixels directly, the entire image is encoded in the parameters θ of the neural network. Given an image I, the coordinate network can be trained by minimizing

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,y)} \| f_\theta(x, y) - I(x, y) \|^2,$$

where (x, y) are sampled pixel coordinates.

Implicit neural representations have several attractive properties:

- They represent signals as **continuous functions**.
- They allow evaluation at **arbitrary resolution**.
- They enable differentiation with respect to spatial coordinates.

Classical Architectures for INRs

Several neural architectures have been proposed for implicit representations.

ReLU Coordinate Networks. One simple approach uses a multilayer perceptron (MLP) with ReLU activation that takes spatial coordinates as input. Although easy to implement, such networks often exhibit a *spectral bias*, meaning they tend to learn low-frequency signals before high-frequency ones.

Fourier Feature Networks [Tancik et al., 2020]. To mitigate spectral bias, Fourier feature mappings are often applied to the input coordinates. A typical mapping is

$$\gamma(x) = [\sin(Bx), \cos(Bx)],$$

where B is a frequency matrix. This allows the network to more effectively represent high-frequency signals.

SIREN Networks [Sitzmann et al., 2020]. SIREN (Sinusoidal Representation Networks) replace standard activations with sinusoidal functions

$$\phi(x) = \sin(\omega x).$$

Such networks are particularly effective for representing signals with rich frequency content and have been widely used in implicit representation tasks.

Meta-Learning Implicit Neural Representations

Traditional INR methods require optimizing a new network for each individual signal (e.g., each image). This can be computationally expensive and does not allow the model to generalize across signals.

To address this, we consider a *meta-learning* approach using a hypernetwork. A hypernetwork is a neural network that generates the parameters of another neural network. In our setting, we train a hypernetwork

$$H_\phi(I) \to \theta,$$

which takes an image I as input and outputs the weights θ of a coordinate network f_θ. The full model therefore consists of two components:

- A hypernetwork H_ϕ that maps an image to network parameters.
- A coordinate network f_θ that represents the image as a continuous function.

During training, the hypernetwork is optimized so that the generated coordinate network reconstructs the input image. The objective is

$$\mathcal{L}(\phi) = \mathbb{E}_I \mathbb{E}_{(x,y)} \| f_{H_\phi(I)}(x, y) - I(x, y) \|^2.$$

This framework allows a single model to generate implicit neural representations for an entire dataset.

Task I: CIFAR-10 Meta-Learning and Downstream Tasks

In the first part of the project, students will train a hypernetwork on the CIFAR-10 dataset. The hypernetwork should take an image I as input and generate the parameters of a coordinate network that represents the image.

For this task, students may use a SIREN-based coordinate network as the implicit representation model. After training the hypernetwork, students should explore whether the learned representations are useful for downstream tasks. For example, the generated network weights or latent features from the hypernetwork can be used as inputs to a classifier for CIFAR-10 image classification. Students should compare different feature representations, such as raw pixel features, hypernetwork latent representations, and generated INR weights. The goal is to investigate whether implicit neural representations capture meaningful semantic information.

Task II: Celeb-A Meta-Learning

In the second part of the project, students will apply the same framework to the Celeb-A dataset, which contains higher-resolution face images.

Students should train a hypernetwork that generates implicit neural representations for Celeb-A images. Compared with CIFAR-10, this task will require handling larger images and potentially more complex representations. Students may experiment with different INR architectures, including SIREN networks, Fourier feature networks, ReLU coordinate networks, and *custom architectures designed by the team.*

Students should compare reconstruction quality, training stability, and computational efficiency across different architectures.

Evaluation Metrics. In addition to the standard Peak Signal-to-Noise Ratio (PSNR), students must evaluate their models using additional perceptual quality metrics, including:

- Structural Similarity Index Measure (SSIM)
- Learned Perceptual Image Patch Similarity (LPIPS)

These metrics capture perceptual similarity between images and are commonly used in modern image generation and reconstruction tasks.

Super-Resolution Testing. Students should also evaluate the super-resolution capability of their implicit neural representations. One possible protocol is:

- Downsample Celeb-A images to a lower resolution (e.g., 32×32 or 64×64).
- Train the hypernetwork to generate INRs using the low-resolution supervision.
- Evaluate the reconstructed images at higher resolutions (e.g., 128×128 or 256×256) by querying the coordinate network on a denser grid.

Students should compare the reconstructed high-resolution images against the original images using PSNR, SSIM, and LPIPS.

Deliverables

Each team should submit the following:

- A complete implementation of the proposed models.
- A project report (6–8 pages) describing the methodology, experiments, and findings.
- Experimental results including visual reconstructions and quantitative metrics.

The report should clearly discuss the architecture choices, experimental setup, and insights obtained from the experiments. The report should also analyze how well implicit neural representations generalize to higher resolutions and whether different INR architectures influence super-resolution performance.